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United States Patent	12409425
Kind Code	B2
Date of Patent	September 09, 2025
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Machine and method to automatically prepare fluid coloring products

Abstract

A machine (**10**) to automatically prepare a fluid coloring product inside a container (**15**), which can already contain a base product, comprises a loading station (**26**), a perforating device to make an aperture in the upper part of the container (**15**), a delivery device (**21**) configured to deliver coloring substances inside the container through such aperture, and a closing device configured to automatically closed such aperture with a closing element, after the delivery of the coloring substances.

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Appl. No.:	**17/631042**
Filed (or PCT Filed):	**August 10, 2020**
PCT No.:	**PCT/IT2020/050202**
PCT Pub. No.:	**WO2021/028957**
PCT Pub. Date:	February 18, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220314176 A1	Oct. 06, 2022

Foreign Application Priority Data

IT	102019000014610	Aug. 12, 2019
IT	102019000014613	Aug. 12, 2019

Publication Classification

Int. Cl.: **B01F29/30** (20220101); **B01F29/62** (20220101); **B01F31/20** (20220101); **B01F33/84** (20220101); **B01F35/42** (20220101); **B01F35/71** (20220101); **B01F101/30** (20220101)

U.S. Cl.:

CPC **B01F29/30** (20220101); **B01F29/62** (20220101); **B01F31/201** (20220101); **B01F33/84** (20220101); **B01F35/423** (20220101); **B01F35/7131** (20220101); B01F2101/30 (20220101)

Field of Classification Search

CPC: B01F (29/30); B01F (29/62); B01F (31/201); B01F (33/84); B01F (35/423); B01F (35/7131); B01F (2101/30); B01F (31/20)

USPC: 366/110; 366/111; 366/114; 366/208-211; 366/219; 366/605; 366/217

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention concerns a machine and corresponding method to automatically prepare fluid coloring products, for example coloring liquids, paints, enamels, inks, or other fluid coloring substances, contained in a container in which, possibly, a base product can already be present. The present invention allows the user to introduce into the machine a container, possibly containing the base product, and to select the color he/she wants to obtain, after which the machine is able to automatically make both an aperture in the upper part of the container, and to deliver the coloring pigments in the container, and also to close said aperture with a suitable closing element, for example a lid, and also to mix the coloring pigments with the base product, in order to obtain the selected coloring product and make it available to the end user, ready for use.

BACKGROUND OF THE INVENTION

(2) Machines for dispensing or delivering coloring products are known, able to introduce a metered quantity of coloring substances, for example coloring pigments, into a container, in which, possibly, a base product is already contained, in order to obtain the desired coloring product.

(3) Mixing machines are also known, often distinct from dispensing machines, able to mix the coloring product obtained using dispensing machines, so as to mix and homogenize the pigments that color the base product and thus obtain a finished product, ready for use.

(4) From the international patent application WO-A-2011/161532 and from the European patent application EP-A1-2.384.808 machines are also known in which there are both means to deliver the coloring product inside a container and also means to mix the product dispensed into the container. In both these known machines, each container is already provided, in its upper part, with a hole which is permanently closed by a membrane of elastic/elastomeric material, configured to be perforated by the injection nozzles which are part of the delivery means.

(5) However, these known machines have the disadvantage that they can only be used with particular, non-standard containers, each provided with the membrane as above.

(6) Other types of machines known in the state of the art are suitable, in particular, to be used to automatically fill metal containers, such as for example cylindrical cans. Some known machines of this type are described, for example, in the prior art documents EP-A1-0.779.241, FR-A1-2.867.768, WO-A2-2007/110764, EP-A1-2.272.584. These machines comprise a dispensing unit for the fluid coloring product which is also able to make an opening hole on the upper surface of the containers, and to plug this opening hole once the dispensing has been completed.

(7) One purpose of the present invention is to provide a machine and to perfect the corresponding method to automatically prepare fluid coloring products, inside a container in which, possibly, a base product may already be present, which are simple, fast and reliable, and which are also able to make, before the delivery, an aperture in the upper part of the container, and also to automatically and hermetically close said aperture after delivery and before mixing the products contained in the container.

(8) Another purpose of the present invention is to provide a machine and to perfect the corresponding method to automatically prepare fluid coloring products, which are also able to perform, after the delivery of the coloring substances into the container, a mixing of the products contained in the container, possibly using members of the machine that are intended for other functions, or using members that are intended for mixing but to perform other functions of the machine, for example, but not only, the delivery as above.

(9) The Applicant has devised, tested and embodied the present invention to overcome the

shortcomings of the state of the art and to obtain these and other purposes and advantages.

SUMMARY OF THE INVENTION

(10) The present invention is set forth and characterized in the independent claims. The corresponding dependent claims describe other characteristics of the invention or variants to the main inventive idea.

(11) In accordance with the above purpose, a machine according to the present invention, to automatically prepare a fluid coloring product contained inside a container, comprises delivery means configured to deliver at least one or more coloring substances, possibly together with other non-coloring substances, inside the container through an aperture provided in the upper part of the latter, when it is in a delivery position.

(12) In accordance with one characteristic of the present invention, the machine also comprises perforating means configured to make the aperture, and closing means configured to automatically and hermetically close the aperture with at least one closing element, preferably of the removable type.

(13) In accordance with one characteristic of the present invention, the machine also comprises mixing means configured to shake the container and therefore mix the coloring substances, possibly with the base product, inside the container when the latter is in a mixing position.

(14) In accordance with one characteristic of the present invention, the machine also comprises a support member configured to support the container, both in the delivery position and also in the mixing position. According to some embodiments provided here, the support member comprises a lower plate provided with its own central axis, preferably vertical, which is substantially parallel to, or coinciding with, a first vertical axis, or axis of delivery, around which the delivery means are disposed when the container is in the delivery position, and which on the other hand is substantially parallel to, or coinciding with, an axis of rotation, or first mixing axis, when the container is in the mixing position.

(15) In accordance with one characteristic of the present invention, the delivery means comprise a delivery device disposed vertically in a fixed position above a loading station of the containers, in substantial alignment with the first vertical axis as above.

(16) In accordance with one characteristic of the present invention, the perforating means and the closing means are mounted on a slider slidable along a transverse axis that intersects the first vertical axis as above.

(17) In accordance with one characteristic of the present invention, the perforating means comprise a perforating device having a second vertical axis; the closing means comprise a closing device having a third vertical axis; and the slider is selectively mobile between an inactive position, in which both the second vertical axis and also the third vertical axis are distant from the first vertical axis, and at least a first operative position, in which the second vertical axis is substantially coinciding with the first vertical axis.

(18) In accordance with another characteristic of the present invention, the slider is also selectively mobile between the inactive position as above and a second operative position, in which the third vertical axis is substantially coinciding with the first vertical axis.

(19) In accordance with another characteristic of the present invention, the least one closing element as above comprises a lid, the closing device comprises a mandrel configured to insert the lid in the aperture of the container, and the machine comprises an operating unit comprising removal means configured to remove the lid from a store in which a plurality of lids is stored, and position it on the mandrel.

(20) In accordance with another characteristic of the present invention, the removal means as above comprise a support on which a removal gripper is mounted having two jaws configured to remove one lid at a time from the store;

(21) furthermore, the support is mobile at least between a removal position, in which the removal gripper is in correspondence with the store, and a delivery position, in which the removal gripper is

in correspondence with the mandrel, when the slider is in the first operative position.
(22) In accordance with another characteristic of the present invention, the lower plate as above is associated with transfer means configured to move it selectively in a bi-directional manner, possibly together with the container, between the delivery position and the mixing position as above, along a first longitudinal axis that intersects perpendicularly both the first vertical axis and also the axis of rotation.
(23) In accordance with another characteristic of the present invention, the mixing means as above further comprise an upper contrast member, parallel to the lower plate; furthermore, the lower plate and the upper contrast member are configured to compress the container between them, when the container is in the mixing position.
(24) In accordance with another characteristic of the present invention, the machine also comprises mixing means comprising at least a fixed structure configured to support the lower plate, and also comprising a rotating support on which an upper slider, which rotatably supports the upper contrast member, and a lower slider, which slidably supports the lower plate, are mounted sliding with reciprocal motion toward or away from each other in the same direction parallel to the axis of rotation.
(25) In accordance with another characteristic of the present invention, second command means are provided to command the reciprocal movement of the two upper and lower sliders toward/away from each other, parallel to the first axis of rotation so as to determine the reciprocal movement toward/away from each other of the upper contrast member and the lower plate, parallel to the axis of rotation.
(26) In accordance with another characteristic of the present invention, the machine also comprises programmable command means, for example comprising an electronic processor, configured to drive the perforating means, in order to make the aperture in the upper part of the container, the delivery means in order to deliver the at least one or more coloring substances into the container through the aperture, and the closing means, while the container is in the delivery position, in a manner coordinated with the displacement of the slider along the transverse axis. The programmable command means as above are also configured to drive the mixing means and to control the movement of the lower plate.
(27) In particular, the programmable command means, for example comprising an electronic processor, are also configured to drive first the transfer means, in order to take the container from the delivery position toward a mixing position, then the mixing means, in order to adequately mix the fluid products contained inside the container, and finally once again the transfer means, in order to return the container from the mixing position to the delivery position in the loading station.
(28) In accordance with another characteristic of the present invention, a method to automatically prepare a fluid coloring product inside a container, which possibly already contains a base product, by means of a machine comprising delivery means configured to deliver at least one or more coloring substances, possibly together with other non-coloring substances, inside the container through an aperture made in the upper part of the latter, comprises, when the container is in a delivery position, at least a perforation step in which perforating means are activated in order to make the aperture, a subsequent delivery step in which the delivery means deliver the at least one or more coloring substances inside the container through the aperture, and a subsequent closing step in which closing means are activated in order to automatically and hermetically close the aperture of the container with at least one closing element, preferably of the removable type.
(29) In accordance with one characteristic of the present invention, the method further comprises a mixing step, in which mixing means mix the at least one or more coloring substances, possibly together with the base product and/or other non-coloring substances, in which it is provided that the container is supported by a support member both during the delivery step, when the container is in the delivery position, and also during the mixing step, when the container is in the mixing position.
(30) In accordance with another characteristic of the present invention, the perforation step, the

delivery step and the closing step as above are performed in succession while the container is in the delivery position, in correspondence with a loading station of the containers.
(31) In accordance with one variant embodiment, the method according to the present invention also comprises a transfer step, in which transfer means, after the closing step as above, transfer the support member, together with the container, from the delivery position to the mixing position and, after the mixing step, transfer the support member, together with the container, from the mixing position to the delivery position.
(32) In accordance with another characteristic of the present invention, simultaneously with the delivery step, or the mixing step, there is provided a step of removing the closing element from a store, in which it is provided to drive means to remove the closing element alternatively between an inactive position, in which the removal means are configured to release the closing elements, one at a time, to a mandrel comprised in the closing means, and a removal position in which the removal means are operationally aligned with the store in order to remove from the latter the closing element disposed at the lowest point.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) These and other characteristics of the present invention will become apparent from the following description of a preferred embodiment, given as a non-restrictive example with reference to the attached drawings wherein:
(2) FIG. **1** is a first right front perspective view which schematically shows a machine according to the present invention, including its external cover;
(3) FIG. **2** is a second right front perspective view of the machine of FIG. **1**, but without its external cover and without any container inserted in it;
(4) FIG. **3** is a third right front perspective view of the machine of FIG. **2**, but with a container shown in a loading position;
(5) FIG. **4** is a top view of the machine of FIG. **2**;
(6) FIG. **5** is a front view of the machine of FIG. **3**, that is, shown with a container in the loading position;
(7) FIG. **6** is a right lateral view of the machine of FIG. **2**;
(8) FIG. **7** is a rear view of the machine of FIG. **2**;
(9) FIG. **8** is a front view of a first operating unit of the machine of FIG. **1**, on an enlarged scale and shown in an inactive position;
(10) FIG. **9** is a front view of the first operating unit of FIG. **8**, but shown in a first operative position;
(11) FIG. **10** is a front view of the first operating unit of FIG. **8**, but shown in a second operative position;
(12) FIG. **11** is a rear left perspective view of the first operating unit of FIG. **8**;
(13) FIG. **12** is a perspective view, taken from below and from the left, of a sub-unit, on an enlarged scale, of the first operating unit of FIG. **8**;
(14) FIG. **13** is a partial perspective view, on a further enlarged scale, of a first detail of the sub-unit of FIG. **12**;
(15) FIG. **14** is a partial front view, on a further enlarged scale, of a second detail of the sub-unit of FIG. **12**;
(16) FIG. **15** is a left front perspective view of the first operating unit of FIG. **8**, of a loading plate and of a second operating unit of the machine of FIG. **1**;
(17) FIG. **16** is a front view, on an enlarged scale, of the second operating unit of FIG. **15**;
(18) FIG. **17** is a left side view, on an enlarged scale, of the second operating unit of FIG. **15**;

(19) FIG. **18** is a front right perspective view of a third operating unit of the machine of FIG. **1**;
(20) FIG. **19** is a rear right perspective view of the third operating unit of FIG. **18**;
(21) FIG. **20** is a right lateral view of the third operating unit of FIG. **18**;
(22) FIG. **21** is a front right perspective view of a detail, on an enlarged scale, of the third operating unit of FIG. **18**;
(23) FIG. **22** is a front view, on an enlarged scale, of the third operating unit of FIG. **18**.
(24) We must clarify that in the present description and in the claims the terms high, low, vertical, horizontal, lower, upper, right, left, high and low, with their declinations, have the sole function of better illustrating the present invention with reference to the drawings and must not be in any way used to limit the scope of the invention itself, or the field of protection defined by the claims. For example, by the term horizontal we mean a plane that can be either parallel to the line of the horizon, or inclined, even by several degrees, for example up to 30°, with respect to the latter.

DESCRIPTION OF AN EMBODIMENT OF THE PRESENT INVENTION

(25) With reference to FIGS. **1**, **2** and **3**, a machine **10** to automatically prepare fluid coloring products according to the present invention substantially comprises three compartments **11**, **12** and **13**, respectively first, second and third, which can also be made in an independent, but modular manner and which are mounted integrated with each other and closed by an external cover **14** (FIG. **1**). For example, the fluid coloring products can be coloring liquids, paints, enamels, inks, or other fluid coloring substances, suitable to be contained in a container **15** (FIGS. **3** and **5**) in which, possibly, a base product can already be present. For example, the base product can be a neutral, transparent, or white paint, to which one or more colorants are then added, substantially formed by coloring pigments.
(26) The containers **15** can be either made of metal, for example of thin steel, or tin, or of plastic material, and preferably have a cylindrical shape, although they can have any other shape whatsoever, such as for example parallelepiped, with a square or rectangular base.
(27) Furthermore, each container **15** has an upper part **16** (FIGS. **3** and **14**), made of a material that can be easily perforated, in which an aperture can be made by the machine **10** itself having, for example, the shape of a cylindrical hole **17** (FIG. **14**), as will be described in detail below.
(28) The cylindrical hole **17** is configured to be automatically and hermetically closed, in the same machine **10**, as will be described in detail below, by means of a covering element, which in the example provided here is a lid **18**, for example made of plastic material, preferably of the removable type. According to one variant, not shown in the drawings, the covering element could consist of an adhesive label.
(29) In this embodiment, regardless of the shape and size of the container **15**, the cylindrical hole **17** has a determinate diameter D**1**, which corresponds to the external diameter of the lower part of the lid **18** (FIG. **16**), so that any lid **18** whatsoever, removed from a stack of lids **18**, all identical to each other, can hermetically close each cylindrical hole **17** made on a container **15** being worked. The lid **18** also has both a determinate internal diameter D**2** (FIGS. **11** and **16**), slightly smaller than the external diameter D**1** of its lower part, and also an external diameter D**3** of its upper part.
(30) In the first compartment **12**, which is the furthest to the left in FIGS. from **1** to **5**, and the furthest to the right in FIG. **7**, there is disposed a plurality of tanks **19**, which can be of any known type whatsoever, for example of the type described in European patent EP-B-1.744.826, or which will be developed in the future. Each tank **19** contains a different coloring substance, for example in the form of coloring pigments having a determinate color. In the example provided here, there are twelve tanks **19**, therefore up to twelve different coloring substances can be contained in them, but it must be clear that the number of tanks **19** can also be lower than or greater than twelve.
(31) A plurality of volumetric pumps **20** (FIG. **5**), of any known type whatsoever, for example of the bellows type, are each connected to a corresponding tank **19** in order to selectively convey the coloring substance contained in the latter toward a single delivery device **21** (FIGS. from **2** to **5** and from **8** to **11**) belonging to a first operating unit **22** disposed in the second compartment **12**.

(32) The delivery device **21** can be of any known type whatsoever, for example of the type described in Italian patent IT-B-1.370.076, or which will be developed in the future, and is positioned so that its center lies on a first vertical axis Z**1** (FIGS. **4** and **5**) which perpendicularly intersects a first longitudinal axis X**1** (FIG. **4**).
(33) In the example provided here, the delivery device **21** comprises a plurality of nozzles **23** (FIG. **11**), also of a known type, the number of which is the same as that of the tanks **19** and of the volumetric pumps **20**. Each nozzle **23** is pointed and disposed in a substantially vertical position with the tip facing downward and is connected to a tubular terminal **24**, which in turn is connected to a corresponding volumetric pump **20** by means of a tubular duct of any known type whatsoever and not shown in the drawings.
(34) The delivery device **21** is mounted on a fixed plate **25** (FIGS. **2**, **4** and **5**) disposed horizontally above a station **26** for loading the containers **15**.
(35) The machine **10** comprises a lower plate **27**, for example with a circular shape which develops axial symmetrical around a central axis of its own, which acts as a support element and optionally as a centering member for the container **15** temporarily inserted in the machine **10** in correspondence with the loading station **26**.
(36) In an inactive or loading position, the central axis of the lower plate **27** lies on the first vertical axis Z**1** (FIGS. **5** and **15**). As will be described in detail below, this position coincides with the position of delivery of the coloring substances inside the containers **15**.
(37) In the front part of the loading station **26** there is disposed a first small door **28** (FIG. **1**), which is normally in a closed position, but which can be opened to allow to introduce or remove a container **15**, and then closed again, for example manually.
(38) The first operating unit **22** (FIGS. **5** and from **8** to **15**), in addition to the delivery device **21**, also comprises a perforating device **29**, having the function of selectively making the cylindrical hole **17** in the upper part **16** of the container **15**, and a closing device **30**, having the function of selectively closing the cylindrical hole **17** with a lid **18**, as will be described in detail below.
(39) The perforating device **29** and the closing device **30** are both mounted on a first slider **31** (FIG. **5**), disposed below the fixed plate **25** and slidable parallel thereto along a transverse axis Y, perpendicular to both the first longitudinal axis X**1** (FIG. **4**) and also to the first vertical axis Z**1** (FIG. **5**), between one of the following three positions: an inactive position, shown in FIGS. **5** and **8**, in which the same first slider **31** is completely displaced to the left of the delivery device **21**; a first operative position, shown in FIG. **9**, in which the same first slider **31** is completely displaced to the right, with the perforating device **29** exactly below the delivery device **21**, that is, coaxial to the first vertical axis Z**1**; a second operative position, shown in FIG. **10**, in which the same first slider **31** is in an intermediate zone, with the closing device **30** exactly below the delivery device **21**, that is, coaxial to the first vertical axis Z**1**.
(40) In particular, the first slider **31** is provided with two pairs of sliding blocks **32** (FIG. **15**), slidable on two guide bars **33**, parallel to each other and supported by the fixed plate **25**, and is commanded by a first electric motor **34** (FIGS. from **8** to **11**), mounted on the upper part of the fixed plate **25** and having its own shaft connected to a toothed wheel **35**, constantly meshed with a horizontal rack **36** (FIG. **11**) integral with the first slider **31**.
(41) The perforating device **29** (FIGS. **12** and **13**) comprises a cylindrical tubular support **37** preferably attached to the lower surface of the first slider **31** and having a second vertical axis Z**2** which intersects the transverse axis Y (FIG. **5**). Inside the cylindrical tubular support **37** there is attached a blade **38** (FIGS. **12** and **13**) preferably with an annular shape, coaxial to the second vertical axis Z**2** and shaped so as to define three cutting elements **39**, for example pointed and with the tips facing downward, and preferably disposed angularly offset by 120° with respect to each other. The three cutting elements **39** have a shape such as to hold between them the cut cylindrical part of the upper part **16** of the container **15** when making the cylindrical hole **17**, so that the part itself does not fall into the container **15**.

(42) The cylindrical tubular support **37**, by means of three vertical rods **40** (only two visible in FIG. **13**) which are disposed angularly offset by 120° with respect to each other, supports a first contrast ring **41** (FIG. **12**), also coaxial to the second vertical axis Z**2** and provided with a central hole **42** having a larger diameter than the external diameter of the blade **38**. Three helical springs **43**, supported by the cylindrical tubular support **37** and disposed around the three vertical rods **40**, normally thrust the first contrast ring **41** toward an inactive position, in which the latter has the lower surface slightly lower than the tips of the three cutting elements **39**.
(43) A thruster element **44** is disposed inside the cylindrical tubular support **37** and is axially slidable along the second vertical axis Z**2**, selectively commanded by an actuation mechanism **45**, preferably mounted in a fixed position (on the left in FIG. **8**) on the upper part of the fixed plate **25** and which can be driven for example by a second electric motor **46**.
(44) The thruster element **44** has the function of selectively expelling toward a collection container **47** (FIG. **2**), for example disposed in the lower part of the second compartment **12**, below the actuation mechanism **45**, each cut cylindrical part of the upper part **16** of the container **15**, after the cylindrical hole **17** has been made and the first slider **31** has again reached its inactive position, as will be described in detail below.
(45) A second small door **48** (FIGS. **1** to **4**) is disposed in the front and lower part of the external cover **14** and is normally closed, but it can be opened, for example manually, to allow to remove the cylindrical parts cut by the blade **38** from the collection container **47**.
(46) The closing device **30** (FIGS. **8**, **12** and **14**) is mounted on the lower part of the first slider **31**, coaxially to a third vertical axis Z**3**, which also intersects the transverse axis Y (FIG. **5**) and is positioned at a first fixed distance L**1** (FIG. **8**) from the second vertical axis Z**2** of the perforating device **29**, for example of about 200 mm.
(47) The closing device **30** comprises a mandrel **49** (FIGS. **12** and **14**) having a cylindrical external surface **50** the diameter of which is equal to the internal diameter D**2** of the lid **18**, and a lower end **51** with a substantially truncated cone shape tapered downward.
(48) The mandrel **49** has the function of temporarily holding a lid **18**, in order to then insert it into the cylindrical hole **17** of the container **15**, as will be described in detail below.
(49) The mandrel **49** is vertically slidable with respect to the first slider **31** and is constantly thrust downward by a pair of helical springs **52** which are coaxial to the third vertical axis Z**3**.
(50) The closing device **30** also comprises a second contrast ring **53**, which is also coaxial to the third vertical axis Z**3** and is provided with a central hole **54** having a larger diameter than the external diameter D**3** of the upper part of the lid **18**. Three helical springs **55** (FIG. **12**) are disposed around three corresponding vertical rods **56**, supported by the mandrel **49** and disposed offset by 120° from each other, and normally thrust the second contrast ring **53** toward an inactive position, in which the latter has the lower surface slightly lower than the lower end **51** of the mandrel **49**.
(51) In the right part of the second compartment **12** there is a second operating unit **60** (FIG. **5**), which has the function of removing one lid **18** at a time from a store **61**, disposed along a fourth vertical axis Z**4**, which also intersects the transverse axis Y (FIG. **5**), and positioning it in a removal position, so that it can be removed by the closing device **30**, as will be described in detail below.
(52) The second operating unit **60** (FIGS. **5**, **15**, **16** and **17**) comprises a second slider **62**, which is slidable vertically along a fifth vertical axis Z**5**, which also intersects the transverse axis Y (FIG. **5**) and is disposed at a second distance L**2** (FIG. **16**), for example of about 100 mm, from the fourth vertical axis Z**4**, by means of a third electric motor **63** (FIG. **15**) mounted on the upper part of the fixed plate **25** and having its own shaft connected to a toothed wheel **64**, constantly meshed with a vertical rack **65** integral with the second slider **62**.
(53) A fourth electric motor **66** (FIG. **16**) is mounted on the second slider **62**, which has its shaft coaxial to the fifth vertical axis Z**5** and on which there is mounted an L-shaped support **67**, on the vertical arm of which a plate **68** is mounted slidable vertically, on which there are pivoted two vertical jaws **69** of a removal gripper **70** (FIGS. **15**, **16** and **17**), configured to remove one lid **18** at

a time from the store **61**, that is, the one at the lowest point.
(54) The distance of the two jaws **69** from the fifth vertical axis Z**5** is equal to the second distance L**2** (FIG. **16**).
(55) The fourth electric motor **66** can be selectively driven in order to take, with a rotation of 180°, the jaws **69** from an inactive position, shown in the drawings, to a removal position, in which they are disposed along the fourth vertical axis Z**4**, that is, exactly aligned with the store **61** of the lids **18**, in order to remove the lowest of the latter.
(56) Furthermore, the plate **68** is vertically mobile between a lowered position and a raised position, by means of a fifth electric motor **71** mounted horizontally on the vertical arm of the support **67** and having its own shaft connected to an eccentric pin **72** (FIG. **17**) inserted in a horizontal slot **73** of the same plate **68**.
(57) The two jaws **69** can be selectively actuated, between an open position, in which they are distanced, or open, and a gripping position, in which they are thrust against each other in order to grip a lid **18**, by means of a sixth electric motor **74** (FIG. **16**) mounted horizontally on the plate **68** and having its own shaft connected to a support element **75** (FIG. **17**) at the two ends of which two small rolls **76** are rotatably mounted, which are equidistant from the axis of the sixth electric motor **74** and which each contact one of the two jaws **69**.
(58) The store **61** (FIGS. **5**, **8** and **15**) comprises vertical guide bars **77** configured so as to allow a plurality of lids **18** to be stacked on top of each other and thus form a stack of lids **18**, with the lowest one positioned at such a level that it can be removed by the jaws **69** of the removal gripper **70**, when these are in their removal position.
(59) On the upper part of the external cover **14** (FIG. **1**) of the machine **10**, exactly above the store **61**, there is a third small door **78** (FIGS. **1**, **2** and **6**), which can be driven manually, to allow to load, when necessary, lids **18** into the same store **61**.
(60) The machine **10** also comprises a mixing device **80** (FIGS. **3**, **4**, **6**, **18**, **19**, **20** and **22**), of the gyroscopic type, such as for example of the type described in international patent application WO-A-2006/008590, which is disposed in the third compartment **13** and is configured to mix the coloring product, or the coloring products, contained in the container **15**, after they have been delivered by the delivery device **21** and after the cylindrical hole **17** has been closed with a lid **18** by the closing device **30**.
(61) In other embodiments, not shown here, but which can be easily understood by a person of skill in the art, the mixing device **80**, instead of the gyroscopic type, could be of any other known type whatsoever, or which will be developed in the future, for example of the vibrational type, such as for example that described in international patent application WO-A-2011/161532 mentioned above, or of the vortex or centrifugal type.
(62) In the embodiment described here, the mixing device **80** is able to simultaneously impart upon the container **15** both a main rotation with respect to a second longitudinal axis X**2** (FIGS. **6**, **7**, **18**, **19** and **20**), parallel to the first longitudinal axis X**1**, and also a secondary rotation with respect to an axis of rotation Z**6**, perpendicular to the second longitudinal axis X**2** and which is defined here also as first mixing axis. In an inactive condition (FIG. **20**) the axis of rotation Z**6** is in a vertical position and also intersects the first longitudinal axis X**1**.
(63) In the embodiment described here, the mixing device **80** (FIGS. **3**, **4**, **6**, **18**, **19**, **20** and **22**) comprises a first structure **81**, which is preferably fixed, on which there is mounted a rotating support **82** rotatable around the second longitudinal axis X**2**. On the front part of the rotating support **82** (on the left in FIG. **20**) there are mounted slidable, in reciprocally opposite senses, a third slider, or upper slider, **83** and a fourth slider, or lower slider, **84**, which are configured to be constantly equidistant from the second longitudinal axis X**2**.
(64) The upper slider **83** has an arm **85** which is protruding toward the second compartment **12** and has an end **86** on which an upper plate **87** is mounted in a rotating manner, parallel to the lower plate **27** and constantly coaxial to the axis of rotation Z**6**.

(65) The first structure **81** comprises an upper plate **88** disposed horizontally, on which a seventh electric motor **89** is mounted in a fixed position, which is configured to reciprocally move the upper **83** and lower **84** slider toward and/or away from each other, with respect to the second longitudinal axis X**2**, and therefore the upper plate **87** and the lower plate **27**, in particular when the latter is in a mixing position, in which its central axis is coaxial to the axis of rotation Z**6**, but also when the lower plate **27** is in the delivery position, in which its central axis is instead coaxial to the vertical axis Z**1**, as will be described in detail below.

(66) On the upper plate **88** there is also mounted an actuation mechanism **90** (FIGS. **18**, **19**, **20** and **22**), which can be of any known type whatsoever, or which will be developed in the future and which for the sake of brevity is not described in detail, which, when the rotating support **82** is stationary in a vertical inactive position, which is shown in the drawings, can be selectively actuated in order to make a mechanical connection between the seventh electric motor **89** and the upper **83** and lower **84** slider in order to make happen their reciprocal movement toward or away from each other. When, on the other hand, the actuation mechanism **90** is deactivated, the rotating support **82**, together with the upper **87** and lower **27** plate, is free to rotate around the second longitudinal axis X**2**, as will be described in detail below.

(67) Furthermore, according to another characteristic aspect of the present invention, when a container **15** to be mixed is disposed on the lower plate **27** (FIGS. **18**, **19**, **20** and **22**), and together they are coaxial to the axis of rotation Z**6**, the same lower plate **27** becomes part of the mixing device **80**, since it also acts as a base element to take the container **15** against the upper plate **87** and compress it, as if in a vice, while the same container **15** is made to rotate in order to mix the coloring products contained inside it.

(68) In order to achieve this, the lower plate **27** is mounted on a fifth slider **91** so that it can always rotate freely around its vertical axis of rotation. In fact, the fifth slider **91** is slidable horizontally along the first longitudinal axis X**1** between a loading position, shown in the drawings, in particular in FIG. **20**, in which the same lower plate **27** is coaxial to the first vertical axis Z**1**, and a mixing position, schematized with a dashed line in FIG. **20**, in which the same lower plate **27** is coaxial to the axis of rotation Z**6**.

(69) The fifth slider **91** is provided with two horizontal bars **92** (FIGS. **18**, **19** and **20**) slidably inserted in two horizontal guide bushings **93** supported by the lower slider **84**.

(70) The mixing device **80** also comprises an eighth electric motor **94** mounted on the lower part of the third compartment **13** (FIG. **7**) and configured to make happen the gyroscopic movement of the upper **87** and lower **27** plate and of the possible container **15** disposed between them. In particular, the eighth electric motor **94**, by means of a transmission belt **94**A, is able to make a pulley **94**B rotate having the hub rotatable on the first structure **81**, coaxially to the second longitudinal axis X**2**, and integral with the rotating support **82**.

(71) Moreover, inside the rotating support **82** there is disposed a transmission, or return, mechanism **95**, which can also be of any known type whatsoever, or which will be developed in the future and which for the sake of brevity is not described in detail, which is able to make the upper plate **87** rotate around the axis of rotation Z**6**, exploiting the rotation of the pulley **94**B.

(72) The selective movement of the fifth slider **91** between its loading position and the mixing position, parallel to the first longitudinal axis X**1**, is obtained by means of a ninth and a tenth electric motor **96** and **97** (FIGS. **21** and **22**) which are mounted on a second structure **98**, which is preferably also fixed, disposed in the lower part of the machine **10**, with their shafts disposed vertically.

(73) Each of the two motors **96** and **97** has its shaft keyed onto a corresponding toothed wheel **99** and **100** respectively, in turn meshed with a rack **101** and **102**, respectively, disposed horizontally. The two racks **101** and **102** are attached to the internal surfaces of two vertical walls **103** and **104** of a sixth slider **105** (FIGS. **20** and **22**), which is slidable horizontally by means of sliding blocks **106** (FIG. **22**) with respect to two guides **107** attached to the external surfaces of the second

structure **98**.
(74) On the upper part of the sixth slider **105** there is attached a connection member **108** which is in engagement with the fifth slider **91**, so that to each horizontal displacement of the same sixth slider **105** (FIG. **20**) parallel to the first longitudinal axis X**1**, there corresponds a similar horizontal displacement of the fifth slider **91** and, therefore, of the lower plate **27**.
(75) Between the second compartment **12** and the third compartment **13** (FIGS. **3**, **4** and **6**) there is disposed a dividing door **109**, parallel to the transverse axis Y and symmetrical with respect to the first longitudinal axis X**1**, which is slidable vertically on a guide frame **110**, having substantially the shape of an inverted U.
(76) The dividing door **109** is normally lowered, in a closed position, in order to isolate the delivery zone, which is located below the delivery device **21**, from the mixing device **80**, and it can be raised, in an open operative position, by means of an eleventh and a twelfth electric motor **111** and **112** (FIGS. **4** and **11**) mounted on the upper part of the fixed plate **25**.
(77) Each of the two motors **111** and **112** has its shaft keyed onto a corresponding toothed wheel **113** and **114**, respectively, in turn meshed with a rack **115** and **116** (FIG. **4**), respectively, disposed horizontally. The two racks **115** and **116** are attached to the lateral flanks of the dividing door **109**.
(78) The machine **10** also comprises an electronic processor **117** (FIGS. **2**, **3**, **6** and **7**) configured and programmed to selectively control and command the delivery device **21**, as well as the perforating device **29**, and the closing device **30**, and also the mixing device **80**, including all twelve electric motors **34**, **46**, **63**, **66**, **71**, **74**, **89**, **94**, **96**, **97**, **111** and **112**, with which position sensors of a known type and not shown in the drawings are associated, which send corresponding signals to the same electronic processor **117**. The machine **10** can also be provided with data input means, of any type whatsoever and not shown in the drawings, by means of which a user can select both the color as well as the hue of the final product he/she wants to obtain.
(79) The functioning of the machine **10** described heretofore, which substantially corresponds to the method according to the present invention to automatically prepare fluid coloring products according to the present invention, is as follows.
(80) In an initial inactive, or loading, position shown in the drawings, the machine **10** has the lower plate **27** lowered and in the loading station **26** (FIG. **2**), coaxial to the first vertical axis Z**1**. Consequently, the upper plate **87** of the mixing device **80** is in its raised position. The first slider **31** (FIG. **8**) is displaced to the left, that is, away from the delivery device **21**, the dividing door **109** (FIG. **3**) is lowered and the three small doors **28**, **48** and **78** are closed. The removal gripper **70** (FIGS. **15** and **16**) faces toward the delivery device **21**, in a position of delivery of a lid **18**, it is in its lowered position and has two of its jaws **69** in the open position.
(81) The method to automatically prepare a fluid coloring product according to the present invention, using the machine **10** described heretofore, comprises a loading step, during which a user, after having opened the first small door **28** (FIG. **1**), either manually or automatically, manually inserts a container **15** (FIG. **3**) on the lower plate **27** of the loading station **26** and positions it so that the container **15** itself is coaxial to the first vertical axis Z**1**.
(82) It should be noted that in this embodiment the container **15** already contains inside it the base substance from which to obtain the desired coloring product, by adding coloring substances. Furthermore, the upper part **16** of the container **15** is devoid of any aperture whatsoever. In fact, the cylindrical hole **17** will be made automatically by the perforating device **29**, in a subsequent perforation step.
(83) The first small door **28** is then closed again, either manually or automatically, and will remain clamped in this position until the completion of the operating cycle, that is, when the container **15**, hermetically closed by a lid **18**, and after having undergone a mixing step, can be removed from the loading station **26**.
(84) Optionally, the machine **10** can automatically carry out a centering operation of the container **15** with respect to the lower plate **27**, by any known mean whatsoever, such as for example those

described in international patent application WO-A-2011/161532 mentioned above, or which will be developed in the future.

(85) In this way, the container **15** is positioned exactly below the delivery device **21**, but still lowered with respect to it.

(86) A perforation step is then carried out to make the cylindrical hole **17** in the upper part **16** of the container **15**. In order to do this, the electronic processor **117** commands the first electric motor **34** in order to translate the first slider **31** to the right, taking it from the inactive position, shown in FIG. **8**, to the first operative position, shown in FIG. **9**, in which the vertical axes Z**1** and Z**2** coincide.

(87) Keeping the first slider **31** stationary in this first operative position, the electronic processor **117** first commands the actuation mechanism **90** (FIGS. **20** and **22**) and then the seventh electric motor **89** in order to cause the raising of the lower slider **84** (FIGS. **15**, **18**, **19**, **20** and **22**) and, consequently, also of the lower plate **27**, until the container **15** is taken against the perforating device **29** (FIG. **9**), until the blade **38** (FIG. **13**) of the latter penetrates the upper part **16** (FIG. **14**) of the same container **15** with its cutting elements **39** and makes the circular hole **17**. A similar downward movement, which in this step is loadless, of the upper plate **87** of the mixing device **80** corresponds to this upward movement of the lower plate **27**.

(88) In addition and simultaneously, while the first slider **31** remains in this first operative position (FIG. **9**), on the mandrel **49** of the closing device **30** there is positioned a lid **18**, which had been previously removed from the store **61**, with a removal step during a previous operating cycle, by driving the second operating unit **60**, and which is gripped between the jaws **69** of the removal gripper **70**, as will be described in detail below. In particular, first the fifth electric motor **71** is commanded, in order to raise the plate **68** and the removal gripper **70**, until the inside of the lid **18**, gripped between the jaws **69**, is taken into contact with the cylindrical external surface **50** (FIG. **12**) of the mandrel **49**, then the sixth electric motor **74** is commanded, in order to open the jaws **69** and free the lid **18**, then once again the fifth electric motor **71** is commanded, in order to lower the plate **68** and the gripper **70**, keeping the jaws **69** open.

(89) Once the cylindrical hole **17** has been made and the lid **18** has been inserted on the mandrel **49** (FIG. **14**), with a reverse command to the seventh electric motor **89** (FIGS. **20** and **22**), the lower plate **27** and the container **15** are lowered again, while the blade **38** temporarily holds the removed part of the upper part **16** (FIG. **14**) of the container **15** between its cutting elements **39** (FIG. **13**). Then the first electric motor **34** is commanded in reverse so that the first slider **31** returns to its initial position, shown in FIG. **8**.

(90) To this downward movement of the lower plate **27** there corresponds a similar upward movement of the upper plate **87** of the mixing device **80**, which thus returns to its initial position.

(91) A delivery step is then carried out, which occurs automatically under the control of the electronic processor **117**.

(92) This delivery step provides that the seventh electric motor **89** (FIGS. **20** and **22**) is commanded again, so that the lower plate **27** and the container **15** mounted thereon are raised again until the central hole **17** (FIG. **14**), made in the upper part **16** of the container **15**, is taken close to the lower tips of the nozzles **23** (FIG. **11**) of the delivery device **21**.

(93) The electronic processor **117** then commands, in a programmed manner, the actual delivery of determinate quantities of at least one or more coloring substances selected automatically, possibly together with other non-coloring substances, in order to obtain the desired color, removing them from the respective tanks **19** (FIG. **2**), by means of the volumetric pumps **20**, and conveying them through the tubular terminals **24** and the nozzles **23** (FIG. **11**) into the container **15**.

(94) At the end of the delivery step, a step is carried out of hermetically closing the cylindrical hole **17** of the container **15** by means of the lid **18** which is already positioned on the mandrel **49**, as previously described. In particular, the electronic processor **117** first possibly once again commands the seventh electric motor **89** to cause the lowering of the lower plate **27**, in order to

move the container **15** away from the nozzles **23** and then, in any case, once again commands the first electric motor **34** so as to translate the first slider **31** to the right, taking it from the inactive position, shown in FIG. **8**, to the second operative position, shown in FIG. **10**, in which the vertical axes Z**1** and Z**3** coincide.

(95) Keeping the first slider **31** stationary in this second operative position, the electronic processor **117** once again commands the seventh electric motor **89** in order to cause the raising of the lower slider **84** (FIGS. **15**, **18**, **19**, **20** and **22**) and, consequently, also of the lower plate **27**, until the container **15** is taken against the lid **18** positioned on the mandrel **49**, until the same lid **18** enters the cylindrical hole **17**, closing it hermetically. Also in this case, a similar downward movement, which also in this step is loadless, of the upper plate **87** of the mixing device **80** corresponds to the upward movement of the lower plate **27**.

(96) Upon completion of the closing step, the electronic processor **117** once again commands the seventh electric motor **89** to cause the lowering of the lower slider **84** (FIGS. **15**, **18**, **19**, **20** and **22**) and, consequently, also of the lower plate **27** and then once again commands the first electric motor **34** so as to translate the first slider **31** toward the left, taking it from its inactive position, shown in FIG. **8**.

(97) Once the cylindrical hole **17** of the container **15** has been hermetically closed with the lid **18**, the electronic processor **117** possibly commands a step of mixing or shaking by means of the mixing device **80**.

(98) In particular, the electronic processor **117** first commands the opening, that is, the raising, of the dividing door **109** (FIGS. **3** and **4**), simultaneously driving the eleventh electric motor **111** and the twelfth electric motor **112** (FIGS. **4** and **11**), and then the horizontal displacement toward the right, parallel to the first longitudinal axis X**1**, of the sixth slider **105** (FIG. **20**), simultaneously driving the ninth electric motor **96** (FIGS. **21** and **22**) and the tenth electric motor **97**. In this way, the fourth slider **91** is also displaced toward the right, together with the lower plate **27** and the container **15** resting on it, until the axis of rotation of the lower plate **27** coincides with the axis of rotation Z**6**, in perfect alignment with the upper plate **87** (a position not shown in the drawings, but easily understood by a person of skill in the art).

(99) Then the electronic processor **117** once again commands the seventh electric motor **89** (FIGS. **15**, **18**, **19**, **20** and **22**) to cause the lowering of the upper slider **83** and the simultaneous raising of the lower slider **84** and, consequently, also of the lower plate **27**, until the container **15** is clamped between the lower plate **27** and the upper plate **87**.

(100) Then the eighth electric motor **94** is driven, which rotates the whole rotating system of the mixing device **80**, that is, the rotating support **82** and, by means of the transmission mechanism **95**, also the upper plate **87** around its axis of rotation Z**6** and consequently also the container **15** and the lower plate **27**.

(101) The actual mixing step consists of a frenetic rotation, in a gyroscopic manner, of the container **15** clamped between the lower plate **27** and the upper plate **87**, for a determinate period of time, for example comprised between 0.5 min and 5 min, in order to thus mix the base product with the coloring substances injected by the delivery device **21** and obtain the desired finished product.

(102) At the end of the actual mixing step as above, the electronic processor **117** stops the eighth electric motor **94** when the rotating support **82** is again in its initial inactive position, with the axis of rotation Z**6** vertical (FIG. **20**), after which it commands first the actuation mechanism **90** and then the seventh electric motor **89**, in the opposite sense to that last performed, in order to cause the raising of the upper slider **83** and the simultaneous lowering of the lower slider **84**, the lower plate **27** and the container **15**.

(103) Then the electronic processor **117** commands the electric motors **96**, **97** in the opposite sense, so that the lower plate **27** and the container **15**, with its content already mixed, return to their initial positions, that is, with the latter coaxial to the first vertical axis Z**1**.

(104) Furthermore, every time the lower plate **27** moves from its loading position, in which it is coaxial to the first vertical axis Z**1**, to its mixing position, in which it is coaxial to the axis of rotation Z**6**, the electronic processor **117** selectively commands the eleventh **111** and twelfth **112** electric motor in order to raise and then once again lower the dividing door **109**.
(105) At the same time as the delivery step, or the mixing step as above, or separately with respect thereto, there is also performed a step of removing a lid **18** from the store **61** by means of the second operating unit **60** (FIGS. **15**, **16** and **17**). In particular, the electronic processor **117** first commands the fourth electric motor **66** in order to rotate the support **67** by 180° and thus take the removal gripper **70**, with the jaws **69** still open, below the store **61**, into a removal position, in alignment with the fourth vertical axis Z**4**. Then the electronic processor **117** first commands the fifth electric motor **71**, in order to raise the plate **68** until it takes the upper part of the two jaws **69** into alignment with the upper part of the lid **18** which is at the bottom of the stack in the store **61**, and then the sixth electric motor **74**, in order to close the jaws **69** so that they clamp the lid **18** between them.
(106) Then the electronic processor **117** commands, in sequence and in the opposite sense to the previous one, first the fifth electric motor **71** in order to lower the plate **68** and the removal gripper **70** to their initial vertical position, with the jaws **69** closed around the lid **18** removed, and then the fourth electric motor **66** in order to rotate the support **67** by 180° and thus take the removal gripper **70** once again to its initial inactive position, shown in FIG. **16**, but with a lid **18** clamped between the jaws **69**, ready to be inserted, in a subsequent operating cycle, in the mandrel **49**, as has been previously described.
(107) At the same time as the removal step as above, or separately from it, an expulsion step is also carried out, during which the part removed from the upper part **16** of the lid **18**, which is still between the cutting elements **39** of the blade **38**, now in alignment with the second vertical axis Z**2**, is thrust toward the collection container **47** below (FIG. **2**) by the thruster element **44** (FIG. **13**) which is lowered by means of the actuation mechanism **45** (FIGS. **2** and **11**), commanded by the second electric motor **46** (FIG. **11**).
(108) At this point the operating cycle is completed, the first small door **28** can be reopened and the container **15**, containing the desired colored product and hermetically closed with the lid **18**, can be removed manually by the user.
(109) It is therefore clear that all the steps of the operating cycle, including the making of the cylindrical hole **17** in the upper part of the container **15** and the hermetic closing of the same cylindrical hole **17**, are managed automatically by the electronic processor **117**, with only the intervention of the same user, without needing the presence of a specialized operator.
(110) Furthermore, according to one variant of the present invention, not shown in the drawings, but easily understood by a person of skill in the art, the machine **10** could be without the mixing device **80**, as described above, and instead be provided with a simplified mixing member, directly connected to the lower plate **27**, in order to rotate the latter, together with the container **15**, around the first vertical axis Z**1**, after the completion of both the step of delivering the coloring substances, injected by the delivery device **21**, and also the step of closing the cylindrical hole **17** with the lid **18**, by means of the closing device **30**. For example, during the drive of the simplified mixing member as above, the container **15** could be kept stationary on the underlying lower plate **27** by mechanical, or electromagnetic, means of a known type.
(111) It should be noted that, advantageously, the lower plate **17**, which acts as a support member of the container **15**, is always supported by the first structure **81**, both during the mixing step by means of the mixing device **80**, and also during the delivery step, when the container **15** is associated with the delivery device **21**.
(112) It is clear that modifications and/or additions of parts may be made to the machine **10** and to the corresponding method as described heretofore, without departing from the field and scope of the present invention.

(113) For example, in a simplified version of the machine **10**, the lower plate **27** could be provided with a mandrel with self-centering jaws, of a known type and possibly motorized, which could keep the container **15** stationary in position. In this case, the presence of the upper plate **87** and of the corresponding members for moving the upper **83** and lower **84** slider toward/away from each other would not be necessary.
(114) Furthermore, for example, the delivery device **21** could move parallel to the first longitudinal axis X**1** from an inactive position, in which it is disposed along the first vertical axis Z**1**, to a delivery position, in which it is disposed along the axis of rotation Z**6**, whereby the latter also becomes the axis of delivery.
(115) It is also clear that, although the present invention has been described with reference to a specific example of one embodiment, a person of skill in the art shall certainly be able to achieve many other equivalent forms of machine to automatically prepare fluid coloring products, and of the corresponding methods, having the characteristics as set forth in the claims and hence all coming within the field of protection defined thereby.

Claims

1. A machine to automatically prepare a fluid coloring product inside a container, the container configured to already contain a base product, the machine comprising: at least one nozzle configured to deliver at least one coloring substance inside the container through an aperture provided in the upper part of the container; a blade configured to make the aperture; a closing device configured to automatically and hermetically close the aperture with at least one closing element when the container is in a delivery position; a mixing device configured to shake the container and therefore mix the coloring substances inside the container when the container is in a mixing position; and a support member configured to support the container, both in the delivery position, and also in the mixing position, wherein the support member is configured to remain stationary to maintain the container in the delivery position while the blade and the closing device shift along a transverse axis to form the aperture and then close the aperture, respectively, and wherein the support member is configured to move along a first longitudinal axis to move the container between the delivery position and the mixing position, the container moving along the first longitudinal axis and away from the transverse axis to move from the dispense position to the mixing position.
2. The machine as in claim 1, wherein the support member comprises a lower plate provided with central axis, which is substantially parallel to, or coinciding with, a first vertical axis around which the delivery at least one nozzle is disposed, when the container is in the delivery position, and which is substantially parallel to, or coinciding with, an axis of rotation when the container is in the mixing position.
3. The machine as in claim 2, wherein the at least one nozzle is disposed vertically in a fixed position above a loading station of the container, in substantial alignment with the first vertical axis, and in that the blade and the closing device are mounted on a slider slidable along a transverse axis which intersects the first vertical axis.
4. The machine as in claim 3, wherein the blade has a second vertical axis, in that the closing device has a third vertical axis, and in that the slider is selectively mobile between an inactive position, in which both the second vertical axis and also the third vertical axis are distant from the first vertical axis, at least a first operative position, in which the second vertical axis substantially coincides with the first vertical axis, and a second operative position, in which the third vertical axis substantially coincides with the first vertical axis.
5. The machine as in claim 4, wherein the at least one closing element comprises a lid, in that the closing device comprises a mandrel configured to insert the lid in the aperture of the container, and in that the closing device is further configured to remove the lid from a store in which a plurality of

lids is stored and position the lid on the mandrel.
6. The machine as in claim 4, further comprising a cylindrical tubular support attached to the slider, developing along the second vertical axis, and inside which there is attached the blade, coaxial to the second vertical axis and shaped so as to define three cutting elements, having a shape such as to hold between the three cutting elements the cut cylindrical part of the upper part of the container when making the cylindrical hole, so that the cut cylindrical part does not fall inside the container.
7. The machine as in claim 6, further comprising a thruster element disposed inside the cylindrical tubular support, and axially slidable along the second vertical axis, and selectively commanded by an actuation mechanism to selectively eject toward a collection container each of the cylindrical parts cut from the upper part of the container, when the slider is in the inactive position.
8. The machine as in claim 2, wherein the lower plate is configured to move selectively in a bi-directional manner, along a first longitudinal axis, which intersects perpendicularly both the first vertical axis, and also the first axis of rotation, between the delivery position and the mixing position, and vice versa.
9. The machine as in claim 2, wherein the mixing device further comprises an upper contrast member, parallel to the lower plate, and in that the lower plate and the upper contrast member are configured to compress the container therebetween, when the container is in the mixing position.
10. The machine as in claim 2, wherein the mixing device comprises at least a fixed structure configured to support the lower plate and also comprises a rotating support on which an upper slider, which rotatably supports the upper contrast member, and a lower slider, which slidably supports the lower plate, are mounted sliding with reciprocal motion such that the upper slider and lower slider are configured to move one or both of closer together and further apart parallel to the axis of rotation.
11. The machine as in claim 10, wherein the mixing device comprises command means to command the reciprocal movement of the upper and lower sliders toward/away from each other, parallel to the first axis of rotation so as to determine the reciprocal movement toward/away from each other of the upper contrast member and the lower plate, parallel to the first axis of rotation.
12. A machine to automatically prepare a fluid coloring product inside a container, the container configured to already contain a base product, the machine comprising: a delivery device disposed vertically in a fixed position above a loading station of the container, the delivery device configured to deliver at least one or more coloring substances inside the container through an aperture provided in the upper part of the container; a blade configured to make the aperture; a mandrel configured to automatically and hermetically close the aperture with a lid when the container is in a delivery position by inserting the lid into the aperture of the container; a support on which a removal gripper is mounted and having two jaws configured to remove one lid at a time from a store in which a plurality of the lids are stored, wherein the support is mobile at least between a removal position, in which the removal gripper is in correspondence with the store, and a delivery position, in which the removal gripper is in correspondence with the mandrel; a mixing device configured to shake the container and therefore mix the coloring substances inside the container when the container is in a mixing position; and a lower plate configured to support the container, both in the delivery position, and also in the mixing position.
13. A machine to automatically prepare a fluid coloring product inside a container, the container configured to already contain a base product, the machine comprising: a delivery device disposed vertically in a fixed position above a loading station of the container, the delivery device configured to deliver at least one or more coloring substances inside the container through an aperture provided in the upper part of the container; a blade configured to make the aperture; a closing device configured to automatically and hermetically close the aperture with at least one closing element when the container is in a delivery position by inserting the at least one closing element into the aperture of the container; a mixing device configured to shake the container and therefore mix the coloring substances inside the container when the container is in a mixing

position; and a lower plate configured to support the container, both in the delivery position, and also in the mixing position; wherein the lower plate is provided with a central axis disposed substantially parallel to or coinciding with a first vertical axis around which the delivery device is disposed with the container in the delivery position, and which central axis is one of substantially parallel to and coinciding with an axis of rotation with the container in the mixing position; wherein the delivery device is disposed vertically in a fixed position above a loading station of the container, in substantial alignment with the first vertical axis; wherein the blade and the closing device are mounted on a slider slidable along a transverse axis which intersects the first vertical axis; and wherein a controller is configured to control movement of the lower plate and to drive: the blade, in order to make the aperture in the upper part of the container; the delivery device in order to deliver the at least one or more coloring substances into the container through the aperture; the closing device, while the container is in the delivery position, in a manner coordinated with the displacement of the slider along the transverse axis; and the mixing device.
14. A method to automatically prepare a fluid coloring product inside a container configured to contain a base product inside of the container, by a machine comprising configured to deliver at least one or more coloring substances inside the container through an aperture made in an upper part of the container, and the machine including a support member configured to support the container in both a delivery position and a mixing position, the method comprising: when the container is in the delivery position, perforating the container with a perforating device to make the aperture; delivering, with the container in the delivery position, the at least one or more coloring substances inside the container through the aperture; automatically and hermetically closing the aperture of the container by a closing device applying at least one closing element to the aperture with the container in the delivery position; maintaining the container in a first location along a transverse axis and along a first longitudinal axis during the perforating of the container, the delivering of the at least one coloring substance inside the container, and the automatically and hermetically closing the aperture; mixing the at least one or more coloring substances in the container with the container in the mixing position; wherein the perforating device and the closing device move together along the transverse axis and relative to the container between each of the perforating of the container, the delivering the at least one or more coloring substances inside the container, and the automatically and hermetically closing the aperture; and wherein the container is at the first location along the transverse axis with the container in the mixing position and such that the container is supported by the support member both when the container is in the delivery position, and when the container is in the mixing position.
15. The method as in claim 14, wherein perforating the container, delivering the at least one or more coloring substances inside the container, and automatically and hermetically closing the aperture are performed in succession while the container is in the delivery position, in correspondence with a loading station of the container.
16. The method as in claim 14, further comprising, after automatically and hermetically closing the aperture, transferring the support member, together with the container, from the delivery position to the mixing position and, after mixing the at least one or more coloring substances in the container, transferring the support member, together with the container, from the mixing position to the delivery position.
17. The method as in claim 14, wherein at the same time as at least one of delivering the at least one or more coloring substances inside the container and mixing the at least one or more coloring substances in the container, removing the closing element from a store in which the closing element is provided.